

SAMUEL OKWUSIUNO

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EDUCATION

University of Guelph

Bachelor of Computing, Software Engineering

Guelph, ON

April 2026

EXPERIENCE

Coding Instructor

June 2023 – Present

Zebra Robotics

Brampton, ON

- Designed and delivered hands-on programming curriculum in C, Python, Java, JavaScript, and HTML/CSS for 30 students across camps, focusing on applied web development, robotics, and algorithmic practice.
- Facilitated interactive technical workshops for 150+ students covering AI fundamentals, programming, and robotics through live demos, debugging sessions, and hands-on coding activities.

Undergraduate Researcher

Jan 2025 – Apr 2025

University of Guelph, Student Mental Health Study

Guelph, ON

- Co-authored a peer-reviewed conference proceeding from a faculty-supervised capstone: *Burnout Profiles of Canadian Undergraduate Students Using Latent Profile Analysis* (ICERI 2025).
- Analyzed 1,000+ survey responses to produce 3 actionable recommendations for improving student mental health support in CS programs and presented findings to faculty stakeholders.

PROJECTS

APEX | Next.js, Prisma, SQLite, Playwright, axe-core, Docker, XRPL, Claude API

Mar 2026

- Built a full-stack accessibility scanning platform that automates 36+ WCAG and AODA compliance checks using Playwright and axe-core, with a scoring system to rate site accessibility.
- Containerized the scanning service with Docker and built a structured prompt pipeline using Claude's API to parse axe-core violation JSON, classify WCAG failure types, and return actionable code-level fix suggestions ranked by severity.
- Implemented an XRPL escrow payment flow with automatic payouts on successful scans and refunds on timeout; won 2nd place in the Ripple XRP Challenge and 3rd place in the BFN Venture Challenge at NSBE Hacks 2026, earning \$1.5K+ in prizes.

Student Web Project Gallery | AWS Lambda, API Gateway, DynamoDB, S3, Python, React

Feb 2026

- Architected and deployed a serverless platform on AWS enabling students to publish web projects to unique S3 static-hosted URLs via presigned S3 uploads and ZIP extraction, supporting concurrent multi-user submissions without provisioned servers.
- Built an event-driven backend with 5 Lambda functions handling authentication, project CRUD, deployment pipelines, and lightweight analytics, using DynamoDB and S3 for scalable low-cost infrastructure.

Pedestrian Trajectory Robustness Pipeline | Python, PyTorch, OpenCV, Ultralytics YOLO

Jan 2026

- Built a robustness-evaluation pipeline for pedestrian trajectory prediction on JAAD, injecting realistic input corruptions including dropped frames, bounding-box jitter, and geometric box distortions.
- Benchmarked Context-Aware and SGNetPose+ with ADE/FDE metrics, finding that geometric distortions produced larger performance degradation than jitter, while interpolated frame drops had a smaller effect.
- Integrated YOLO detections into the prediction pipeline using IoU-based matching and temporal interpolation for missing boxes; detector-derived inputs raised ADE by 75.0% for SGNetPose+ and 40.2% for Context-Aware versus ground-truth input.

Housing Index | Spring Boot, React, TypeScript, Python, MariaDB, Docker

metropolitan.foundre.app

- Built a full-stack analytics platform using Spring Boot, React, TypeScript, and MariaDB to aggregate, query, and visualize housing and labour market datasets for Toronto and Hamilton.
- Developed Python ETL pipelines to collect, transform, and load Statistics Canada housing starts, labour force microdata, and annual labour force rates into a relational database.
- Eliminated **225,977 redundant SQL round-trips** (226,000 → 23) by replacing per-row inserts with batched bulk writes, reducing ETL load time from **8+ minutes** to **~2 seconds** for **113,001 labour records**.
- Deployed the application to a VPS using Docker Compose, GitHub Actions, and Caddy, and implemented observability with Prometheus, Grafana, alerting, and external uptime monitoring; maintained **99.981% uptime** over the last 30 days with **1 incident totaling 5m 5s** of downtime.

TECHNICAL SKILLS

Languages: Java, SQL, Python, TypeScript, JavaScript, C, Go, HTML/CSS

Frameworks/Libraries: Spring Boot, React, Next.js, Prisma, Playwright, PyTorch, Pandas, Material UI

Cloud/Tools: AWS, Docker, Git, GitHub Actions, REST APIs, CI/CD, Linux, Caddy

Concepts: Object-Oriented Programming, Data Structures & Algorithms, ETL, Data Pipelines, Data Modeling, Agile/Scrum, System Design